

KUCAMPUS BITE

Change Request Document

Change Request Title:

Use of Twilio Sandbox Instead of WhatsApp Business API

Date of Meeting:

5 October 2025

Facilitator (Project Manager):

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Summary of Discussion:

Business Need and Benefits:

This change was proposed to address the project's budget constraints while still ensuring that the WhatsApp-based communication feature remains functional for demonstration and testing purposes.

Instead of using the paid WhatsApp Business API, the team will integrate **Twilio Sandbox**, a free alternative that allows limited but sufficient messaging functionality.

Additionally, instead of sending messages directly to canteen owners, the system will currently display working message simulations to showcase how the feature will operate in production.

This approach maintains project progress, reduces costs, and enables a realistic system demonstration for stakeholders without requiring additional financial investment.

Technical Feasibility:

The change is technically feasible as Twilio Sandbox offers a straightforward integration process compatible with the system's existing architecture. Only minor configuration updates are needed in the backend API connection.

The simulated messaging workflow ensures the system logic and user interface remain consistent with future production requirements once the paid API is adopted.

Cost Impact:

The change reduces project costs, as it eliminates the need for a paid WhatsApp Business API subscription.

No new hardware or software purchases are required. The use of the free Twilio Sandbox version supports continued feature development within the current budget.

Quality/Testing Impact:

Testing will focus on verifying message simulation, system flow, and API integration behavior under the Twilio Sandbox environment.

Although actual message delivery to canteens will not occur in this phase, the mock simulation allows comprehensive testing of message triggers, responses, and user interactions.

User Experience Impact:

End users (canteen owners) will not receive real-time messages in the current phase. However, the simulated messaging interface will clearly demonstrate how the communication system will function after final deployment.

This ensures stakeholders can visualize the feature's workflow and experience minimal differences once the live API is implemented later.

Risks Identified:

- Limited functionality in Twilio Sandbox compared to the paid WhatsApp Business API.
- Possibility of minor discrepancies during the future transition to the full API.
- Temporary unavailability of real-time user communication until the production API is activated.

Final Decision:

☒ **Approve** ☐ Reject ☐ Defer

Justification/Comments:

The change is approved due to its cost-effectiveness and practicality during the development phase. Using the Twilio Sandbox allows the team to demonstrate system capability without

exceeding the project budget.

This decision ensures continuity of work, avoids financial strain, and provides a clear path for upgrading to the official WhatsApp Business API in future releases.

Next Steps / Action Items:

- Implement message simulation features in the current build to demonstrate functional workflows.
- Prepare a transition plan for switching to the paid WhatsApp Business API once budget approval is available.